

Article

Optimization of Active Distribution Network Operation with SOP Considering Reverse Power Flow

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Abstract: As the penetration of distributed renewable energy increases, the phenomenon of bidirectional power flow in distribution networks becomes increasingly severe. Traditional regulation devices like OLTC (on-load tap changer) and CB (capacitor bank) cannot effectively mitigate reverse power flow in distribution networks due to their limitations. The transmission capacity of the distribution network under reverse power flow is approximately 50% of the rated capacity of the OLTC, leading to issues such as voltage limit violations and high wind and solar curtailment rates. This paper proposes a method for calculating the reverse power flow delivery capacity of distribution networks, quantitatively describing the distribution network's delivery limits for reverse power flow. Based on this, a joint optimization model for multiple distribution networks with an SOP is established. The SOP is utilized to share reverse power flow delivery capacity among multiple distribution networks, enhancing operational economy and increasing the accommodation of the DG. Finally, the method's effectiveness and correctness are verified in the IEEE 33-node system. The results validate that while joint operation does not enhance the reverse flow transmission capacity of a single distribution network, it can, through the shared reverse flow transmission capacity approach, elevate the reverse flow transmission capacity to approximately 70% during the majority of time periods.

Keywords: distribution network; soft open point; power electronic devices; reliability evaluation method; mixed integer linear programming model

Citation: Ma, Z.; Cao, F.
Optimization of Active Distribution
Network Operation with SOP
Considering Reverse Power Flow.
Appl. Sci. **2024**, *14*, 11797. <https://doi.org/10.3390/app142411797>

Academic Editor: Roberto Zivieri

Received: 17 October 2024

Revised: 19 November 2024

Accepted: 6 December 2024

Published: 17 December 2024



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1. Introduction

In recent years, as environmental issues such as global warming become increasingly severe, the distributed generation (DG) of distributed renewable energy sources has rapidly developed [1–3]. The distribution network, as the main carrier of the DG, has seen a continuous increase in the penetration rate of the DG within it [4]. This has highlighted the issue of bidirectional power flow [5] in distribution networks, which can, in severe cases, lead to over-voltage problems [6]. When confronted with significant reverse power flows, traditional regulatory equipment like on-load tap changer (OLTC) and capacitor bank (CB) exhibit slow regulation speeds, an inability to output continuously, and limited operational ranges [7–12]. Consequently, the distribution network's capacity to handle reverse power flow is insufficient, leading to voltage limit violations, wind and solar curtailments, and other issues that hinder the further development of the distribution network.

In addressing the aforementioned challenges, flexible distribution devices, exemplified by soft open point (SOP) [13], have garnered significant attention in recent years. SOPs, composed of fully controllable power electronic devices, are free from the operation frequency limitations of traditional mechanical switches. They allow rapid and precise control [14,15] of active and reactive power on connected feeders, making them an

excellent choice for addressing supply–demand imbalances in distribution networks, thereby greatly enhancing network operational efficiency [16–18]. Considerable research has already been conducted on SOP in distribution network optimization. For instance, the literature [19,20] explores the working principle of SOPs in network optimization and establishes an optimal scheduling model by coordinating various controllable resources, thus achieving optimal power flow distribution. The literature [21] proposes a time-segmented reactive compensation strategy. By joint optimization of an SOP and energy storage systems, targeting minimization of average voltage deviation and network loss, operational efficiency is enhanced. The literature [22] develops a two-stage day-ahead and intra-day optimization model incorporating an SOP and other flexible resources, with distributed generation (DG) penetration as the objective function, improving economic efficiency while boosting DG accommodation. The literature [23] establishes an active–reactive coordination optimization model for the network, resolving voltage limit issues by regulating reactive compensation devices, photovoltaic inverters, SOPs, and other adjustable equipment. Additionally, several studies, including [24], address voltage issues arising from two-way power flow due to increased DG penetration by establishing a bidirectional transformer model to quantify control curves under bidirectional flow and mitigating voltage fluctuations through multi-layer regulation of energy storage devices. However, there is limited research on detailed modeling of reverse power flow transmission capacity in distribution networks and on utilizing SOP flexibility to address challenges of insufficient reverse power flow transmission capability.

Based on the aforementioned issues, this paper's contributions are as follows:

- (1) It analyzes the impact of OLTC parameters, the OLTC's low-voltage side reactive compensation, and DG installation location on the reverse power flow capacity of distribution networks. A refined model is established, enabling precise calculations of reverse power flow capacity considering these influencing factors.
- (2) Addressing the issue of insufficient reverse power flow capacity in distribution networks, a multi-distribution network joint operation optimization model is established based on an SOP. This interconnects multiple distribution networks via an SOP, enhancing the reverse power flow capacity of individual networks, thus improving the overall operational efficiency of the distribution network.
- (3) Using the improved IEEE 33-node system, the paper examines the optimization role of an SOP when introduced in various operational scenarios, impacting the reverse power flow capacity and overall operational efficiency of the distribution network.

2. Analysis of Factors Affecting Reverse Power Flow Transmission Capacity in Distribution Networks

Reverse power flow in a distribution network occurs when the output from the DG exceeds the load demand of the distribution network, causing power to flow backward to the upstream grid. The reverse power flow capacity in distribution networks is influenced by the following three factors: The first factor is the parameters of the OLTC connected between the distribution network and the upstream grid, including the rated capacity of the OLTC, the difference in the per unit values of voltage on both sides of the OLTC, as well as the allowable maximum and minimum tap changer ratios. The second factor is the capacity of reactive power compensation that can be deployed on the low-voltage side of the OLTC. The third factor is the electrical distance from the DG to the low-voltage side of the OLTC.

First, the nameplate-rated capacity of the OLTC directly determines the maximum reverse power flow that can be transmitted. Additionally, the magnitude of reverse power flow during transmission is proportional to the difference in the per unit values of voltage on both sides of the OLTC. Therefore, the greater the difference, the larger the reverse power flow capacity. Since the per unit value of the high-voltage side remains constant and is lower than that of the low-voltage side during operation, with increasing

transmission capacity, the per unit value of the low-voltage side voltage increases. Consequently, the OLTC's transformation ratio also increases. When this ratio reaches its upper limit, further increase in reverse power flow capacity is not possible.

Regarding the reactive power compensation capacity on the low-voltage side of the OLTC, to maintain the power factor of the OLTC primary side (low-voltage side during reverse flow) at the required value, as the reverse power flow capacity increases, more reactive compensation capacity is required on the low-voltage side. Once the full compensation capacity is utilized, further increase in reverse power flow capacity is not possible.

Concerning the electrical distance from the DG to the OLTC's low-voltage side, as the reverse power flow capacity increases, the voltage drop along the line from the DG to OLTC also increases [25]. When this voltage drop reaches its allowable maximum, further increase in transmission capacity necessarily requires reduced allowable electrical distance between the DG and the OLTC to prevent voltage limit violations [26,27]. When the electrical distance is reduced to zero, meaning the DG node coincides with the OLTC's low-voltage side node, any further capacity increase is not possible.

The circuit diagram in Figure 1 shows the equivalent representation of all distributed generation (DG) and loads in the distribution network. Here, $R_T^* + jX_T^*$ denotes the internal impedance of the OLTC, Z_{DG}^* refers to the per unit value of the impedance from the equivalent DG source to the OLTC's low-voltage side, and Z_L^* represents the impedance from the equivalent DG source to the equivalent load. Q_{C1}^* and Q_{C2}^* are the reactive power compensation on the high-voltage and low-voltage sides of the OLTC, respectively. U_1^* , U_2^* , and U_3^* denote the voltage values at the OLTC's high-voltage side, low-voltage side, and equivalent DG nodes, respectively. S_1^* and S_2^* are the magnitudes of the power flow injected into and flowing out of the OLTC under forward power flow conditions. $S_{2,re}^*$ and $S_{1,re}^*$ represent the apparent power injected into and flowing out of the OLTC during reverse power flow conditions. K^* signifies the transformer turns ratio.

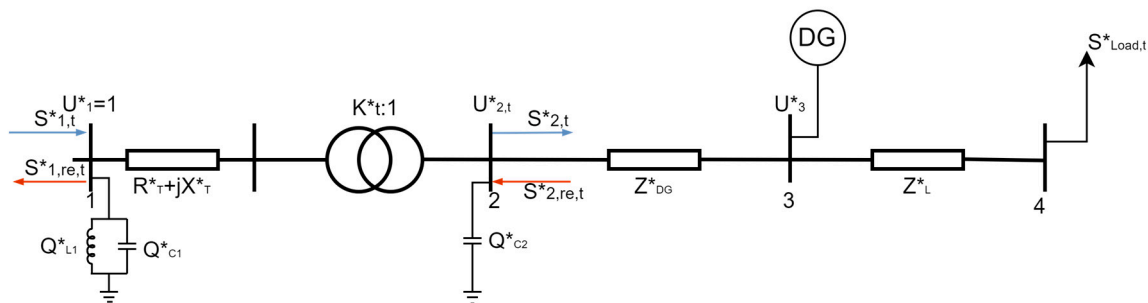


Figure 1. Equivalent circuit diagram of a distribution network.

2.1. Impact of OLTC Parameters on Reverse Power Flow Transmission Capacity

The OLTC's rated capacity determines its transmission capacity limit during power flow, regardless of the flow direction. The transmission capacity is proportional to the per unit voltage difference between the OLTC's high- and low-voltage sides, again irrespective of flow direction. The relationship between the OLTC's transmission capacity under forward and reverse power flows and the per unit voltage difference across both sides is:

$$\begin{aligned}
 dU_t^* &= U_{1,t}^* - U_{2,t}^* = U_{1,t}^* - \sqrt{\left(U_{1,t}^* - \frac{P_{1,t}^* R_T^* + Q_{1,t}^* X_T^*}{U_{1,t}^*} \right)^2 + \left(\frac{P_{1,t}^* X_T^* - Q_{1,t}^* R_T^*}{U_{1,t}^*} \right)^2} \\
 &\approx \frac{P_{1,t}^* R_T^* + Q_{1,t}^* X_T^*}{U_{1,t}^*} = \frac{S_{1,t}^* Z_T^* (\cos \varphi_T \sin \varphi_{Z_T} + \sin \varphi_T \cos \varphi_{Z_T})}{U_{1,t}^*} = \frac{S_{1,t}^* Z_T^* c_T}{U_{1,t}^*}
 \end{aligned} \quad (1)$$

$$dU_{re,t}^* = U_{2,t}^* - U_{1,t}^* = U_{2,t}^* - \sqrt{\left(U_{2,t}^* + \frac{P_{1,t}^* R_T^* + Q_{1,t}^* X_T^*}{U_{2,t}^*}\right)^2 + \left(\frac{P_{1,t}^* X_T^* - Q_{1,t}^* R_T^*}{U_{2,t}^*}\right)^2} \quad (2)$$

$$\approx \frac{P_{1,t}^* R_T^* + Q_{1,t}^* X_T^*}{U_{2,t}^*} = \frac{S_{2,re,t}^* Z_T^* (\cos\varphi_T \sin\varphi_{Z_T} + \sin\varphi_T \cos\varphi_{Z_T})}{U_{2,t}^*} = \frac{S_{2,re,t}^* Z_T^* c_T}{U_{2,t}^*}$$

In this context, $S_{1,t}^*$ and $S_{2,re,t}^*$ represent the per unit values of forward and reverse flow capacities of the OLTC during period t . Their values are less than the rated per unit capacity of the OLTC, i.e., less than 1. $U_{1,t}^*$ and $U_{2,t}^*$ are the per unit values of the high- and low-voltage sides of the OLTC during period t . dU_t^* and $dU_{re,t}^*$ are the differences in the per unit values of voltages on the high and low sides for forward and reverse flows, respectively. Z_T^* is the per unit internal impedance of the OLTC. The power factor $\cos\varphi_T$ is required on the primary side of the OLTC, while $\cos\varphi_{Z_T}$ is the power factor angle of the OLTC internal impedance. It can be observed that regardless of the flow direction, the flow capacity of the OLTC increases as the difference in the per unit voltage values between the high- and low-voltage sides increases.

The high- and low-voltage sides can be represented as per unit values:

$$U_{1,t}^* = \frac{U_{1,t}}{U_{1N}} = 1 \quad (3)$$

$$U_{2,t}^* = \frac{U_{2,t}}{U_{1N}/K_t} = \frac{a_t \cdot U_{2N} \cdot K_t^* \cdot K_N}{U_{1N}} = \frac{b_t \cdot K_t^*}{1.1}, b_t \in (1.00, 1.05] \quad (4)$$

$$K_{min}^* \leq K_t^* \leq K_{max}^* \quad (5)$$

In this context, U_{1N} and U_{2N} are the nominal voltage values at the high- and low-voltage sides, respectively. K_t^* , K_{max}^* , and K_{min}^* denote the per unit tap ratios, maximum tap ratio, and minimum tap ratio for the on-load tap changer (OLTC) at time period t , while K_N refers to the rated tap ratio of the OLTC. Since the OLTC connecting the distribution network and the upper grid is designed to function as a step-down transformer, it follows that $K_N = U_{1N}/1.1U_{2N}$. The coefficient b_t represents the ratio between $U_{2,t}$ and U_{2N} . Under forward power flows, to ensure the voltage at the end of the distribution feeder does not drop below the lower limit, $U_{2,t}$ is generally set to 1.05 times U_{2N} , resulting in $a_t = 1.05$. During reverse power flow, to guarantee $U_{1,t}^* < U_{2,t}^*$, $U_{2,t}$ is chosen between 1.00 to 1.05 times U_{2N} , making $b_t \in (1.00, 1.05]$.

By substituting Equations (3) and (4) into (1) and (2), the relationship between the per unit values of forward and reverse power transmission capacity and K_t^* is obtained as follows:

$$S_{1,t}^* = \frac{\left(1 - \frac{b_t K_t^*}{1.1}\right)}{Z_T^* c_T} \quad (6)$$

$$S_{2,re,K,t}^* = \frac{\left(\frac{b_t K_t^*}{1.1} - 1\right) \frac{b_t K_t^*}{1.1}}{Z_T^* c_T} \quad (7)$$

In expressions (6) and (7), when the forward flow occurs, a smaller K_t^* results in a larger $S_{1,t}^*$; conversely, during reverse flow, a larger K_t^* results in a larger $S_{2,re,t}^*$. This phenomenon arises because the greater the flow transmitted through the OLTC, the larger the difference between the per unit values of the high-voltage and low-voltage side voltages becomes. Assuming $U_{1,t}^*$ always equals 1, as transmission capacity increases, $U_{2,t}^*$ varies more significantly. Given $U_{2,t}^* = b_t K_t^* / 1.1$, where b_t is primarily determined by the operational state of the distribution network, the transmission capacity is thus determined solely by K_t^* . Additionally, the maximum difference in the per unit values of high-voltage

and low-voltage side voltages during forward flow exceeds that during reverse flow, resulting in OLTC transmission capacity being smaller in reverse flow compared to forward flow, as demonstrated below:

$$\begin{aligned} dU_{max}^* - dU_{re,max}^* &= \left(1 - \frac{b_t K_{min}^*}{1.1}\right) - \left(\frac{b_t K_{max}^*}{1.1} - 1\right) \\ &= 2 - \frac{b_t(K_{max}^* + K_{min}^*)}{1.1} \\ &= 2 - \frac{2b_t}{1.1} > 0, b_t \in (1.00, 1.05] \end{aligned} \quad (8)$$

The maximum differences in the per unit voltage values for the OLTC's high- and low-voltage sides during forward and reverse power flow are denoted as dU_{max}^* and $dU_{re,max}^*$, respectively. When K_{max}^* and K_{min}^* are set to 1.1 and 0.9, the relationship between forward and reverse power flow transmission capacity and K_t^* can be observed in Figure 2. The horizontal axis represents the per unit value of the OLTC tap ratio, while the vertical axis represents the per unit value of the OLTC power flow transmission capacity. The blue curve illustrates the variation of $S_{1,t}^*$ with K_t^* when $b_t = 1.05$ (as it is generally assumed that the OLTC's low-voltage side voltage is 1.05 times its rated voltage during operation). The cluster of red curves illustrates the changes in $S_{2,re,t}^*$ with K_t^* as b_t gradually increases from bottom to top. Both $S_{1,t}^*$ and $S_{2,re,t}^*$ must remain greater than zero. In the forward power flow scenario, as K_t^* decreases, $S_{1,t}^*$ reaches the OLTC's rated capacity, achieving a value of 1. However, in the reverse power flow scenario, when both b_t and K_t^* are at their maximum, the maximum value of $S_{2,re,t}^*$ is approximately 84% of the OLTC's rated capacity.

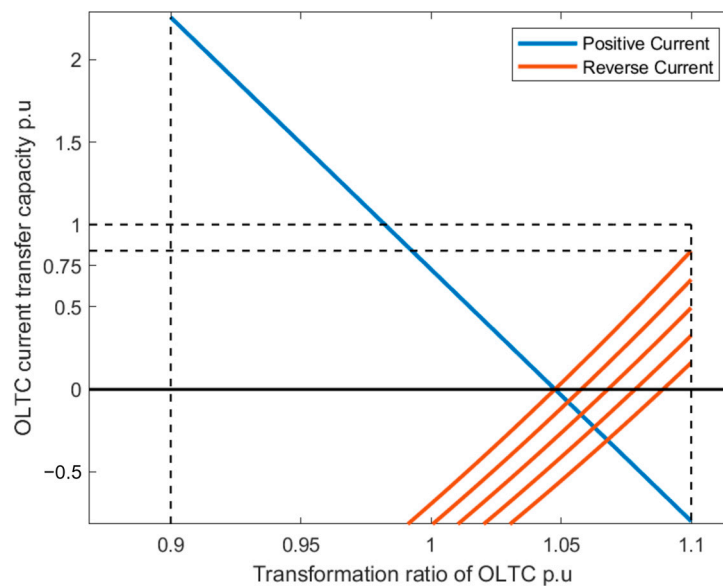


Figure 2. Relationship diagram of the OLTC reverse power transmission capability and the OLTC tap ratio.

2.2. Reactive Power Compensation on the Low-Voltage Side of Oil-Immersed Tap Changers Influences Reverse Power Flow Capacity

During OLTC operation, reactive power compensation must be applied on both sides to maintain power factor. In the presence of forward power flow, the power factor of the upstream grid might not meet the OLTC's primary requirements. Hence, reactive power is added on the high-voltage (primary) side to adjust the power factor. Also, considering the mismatch between the OLTC's low-voltage (secondary) side power factor and the distribution network's power factor, reactive compensation is introduced on the low-voltage

side as well. In reverse power flow scenarios, due to the high-power factor derived from DG units, capacitive reactive power is introduced on the low-voltage (primary) side to lower the power factor. Similarly, as the OLTC's internal reactive losses surpass active losses, capacitive reactive power is added to the high-voltage (secondary) side to maintain the power factor. To facilitate the analysis of changes in active and reactive power, based on Figure 1, this paper employs the tangent value of the power factor angle to describe the aforementioned compensation process. For forward power flow, the tangent values of the power factor angle on both sides of the OLTC can be expressed as follows:

$$\tan\varphi_{1,t} = \frac{Q_{1,t}^*}{P_{1,t}^*} \quad (9)$$

$$\tan\varphi_T = \frac{Q_{1,t}^* - Q_{C1,t}^*}{P_{1,t}^*} \quad (10)$$

$$\tan\varphi_{2,t} = \frac{Q_{1,t}^* - Q_{L1,t}^* - \Delta Q_{T,t}^* + Q_{C2,t}^*}{P_{1,t}^* - \Delta P_{T,t}^*} \quad (11)$$

$$\Delta P_{T,t}^* + j\Delta Q_{T,t}^* = \frac{(P_{1,t}^*)^2(1 + \tan^2\varphi_T)}{(U_{1,t}^*)^2} \cdot (R_T^* + jX_T^*) \quad (12)$$

In forward power flow conditions, $P_{1,t}^*$ and $Q_{1,t}^*$ are the per unit values of active and reactive power injected from the higher-level grid into the high-voltage side. $\varphi_{1,t}$ represents the power factor angle, φ_T is the required power factor angle on the primary side of the OLTC, and $\varphi_{2,t}$ is the required power factor angle on the low-voltage side of the OLTC. $Q_{C1,t}^*$ and $Q_{C2,t}^*$ are the per unit values for reactive compensation capacity on the high- and low-voltage sides. $\Delta P_{T,t}^*$ and $\Delta Q_{T,t}^*$ denote the per unit active and reactive power losses on the OLTC at this time.

Below reverse flow conditions, the tangent of the power factor angle for each node is represented as:

$$\tan\varphi_{2,re,t} = \frac{Q_{2,re,t}^*}{P_{2,re,t}^*} \quad (13)$$

$$\tan\varphi_T = \frac{Q_{2,re,t}^* + Q_{C2,re,t}^*}{P_{2,re,t}^*} \quad (14)$$

$$\tan\varphi_{1,re,t} = \frac{Q_{2,re,t}^* + Q_{C2,re,t}^* - \Delta Q_{T,re,t}^* + Q_{C1,re,t}^*}{P_{2,re,t}^* - \Delta P_{T,re,t}^*} \quad (15)$$

$$\Delta P_{T,re,t}^* + j\Delta Q_{T,re,t}^* = \frac{(P_{2,re,t}^*)^2(1 + \tan^2\varphi_T)}{(U_{2,t}^*)^2} (R_T^* + jX_T^*) \quad (16)$$

$P_{2,re,t}^*$ and $Q_{2,re,t}^*$ represent the per unit active and reactive power injected into the distribution network under reverse power flow at the low-voltage side of the OLTC. $\varphi_{2,re}$ denotes the power factor angle at this moment, while $\varphi_{1,re,t}$ indicates the required power factor angle at the high-voltage side at this time. $Q_{C1,re,t}^*$ and $Q_{C2,re,t}^*$ represent the per unit reactive compensation capacity at the high- and low-voltage sides at this time. $\Delta P_{T,re,t}^*$ and $\Delta Q_{T,re,t}^*$ represent the active and reactive power losses on the OLTC at this moment.

The relationship between the per unit values of power flow transfer capacity and reactive power compensation at both ends in positive and negative sequences is derived after arranging Equation (9) through (16) according to the aforementioned relationships.

$$S_{1,t}^* = \min \left\{ \frac{Q_{L1,t}^*}{(\tan\varphi_{1,t} - \tan\varphi_T)\cos\varphi_{1,t}}, \sqrt{\frac{Q_{C2,t}^*}{(-\tan\varphi_2 R_T^* + X_T^*) \cdot (1 + \tan^2\varphi_T) \cdot \cos^2\varphi_{1,t}}} \right\} \quad (17)$$

$$S_{2,re,t}^* = \min \left\{ \frac{Q_{C2,re,t}^*}{(\tan\varphi_T - \tan\varphi_{2,re,t})\cos\varphi_{2,re,t}}, \sqrt{\frac{(\frac{b_t \cdot K_t^*}{1.1})^2 \cdot Q_{C1,re,t}^*}{(-\tan\varphi_{1,re,t} R_T + X_T)(1 + \tan^2\varphi_T)\cos^2\varphi_{2,re,t}}} \right\} \quad (18)$$

The OLTC reverse power flow limits, $S_{1,t}^*$ and $S_{2,re,t}^*$, represent forward and backward power flow scenarios. Both values increase as the reactive power compensation on the high and low sides increases. When the forward power flow $S_{1,t}^*$ reaches the OLTC's rated capacity (equal to 1), $Q_{L1,t}^*$ and $Q_{C2,t}^*$ denote the initial installation capacity on high and low sides. If the reactive power compensation needed for reverse flow exceeds this, the reverse power limit peaks. Equation (18) seeks the maximum $S_{2,re,t}^*$ and hence uses K_{max}^* . The real distribution network data of $\tan\varphi_{1,t}$, $\tan\varphi_T$, $\tan\varphi_{2,t}$, $\tan\varphi_{1,re,t}$, and $\tan\varphi_{2,re,t}$ show the relationship between power flow limits and reactive compensation, as depicted in Figure 3.

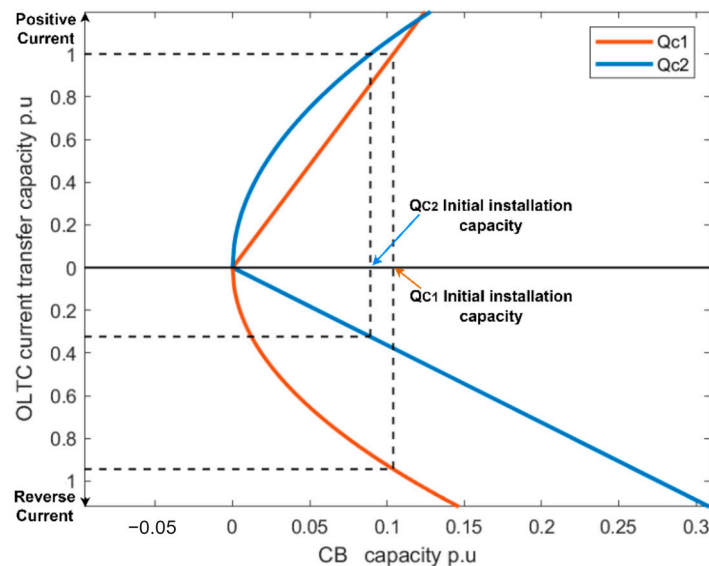


Figure 3. The relationship between OLTC power flow capacity and reactive power compensation.

In the diagram, the vertical axis represents the forward and reverse power flow capacities of the OLTC, while the horizontal axis indicates reactive compensation capacities on both sides of the OLTC, with the red and blue curves representing high-voltage and low-voltage sides, respectively. When $S_{1,t}^*$ equals 1, the x-coordinate values of the red and blue curves denote the initial reactive compensation installed capacities on the high- and low-voltage sides of the OLTC. Under these installed capacities, the maximum reverse power flow capacity that the low-voltage side can compensate is about 33% of the OLTC's rated capacity, whereas the high-voltage side can compensate up to approximately 95%. This indicates that the reverse power flow capacity is primarily constrained by the reactive compensation capacity on the low-voltage side of the OLTC. Therefore, Equation (18) can be expressed as:

$$S_{2,re,t}^* = \frac{Q_{C2,re,t}^*}{(\tan\varphi_T - \tan\varphi_{2,re,t})\cos\varphi_{2,re,t}} \quad (19)$$

To enhance the reverse power flow capacity, increase the reactive power compensation installation capacity on the low-voltage side. After increasing the installation capacity, the relationship between $S_{2,re,Q,t}^*$ and $Q_{C2,re,t}^*$ can be seen in Figure 4.

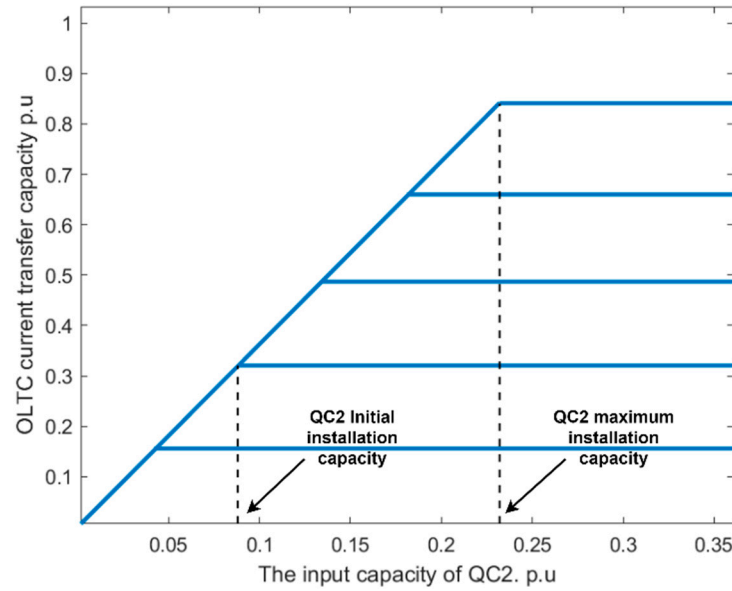


Figure 4. Reverse power flow involves understanding the relationship between reactive power compensation capacity on the low-voltage side.

The graph's horizontal axis represents the reactive power compensation capacity on the OLTC's low-voltage side, while the vertical axis indicates the reverse power flow transfer capacity. One can observe that $S_{2,re,t}^*$ increases with the rise in $Q_{C2,re,t}^*$. The horizontal lines from bottom to top signify the maximum $S_{2,re,t}^*$ achievable with increasing b_t , reflecting the constraint imposed by K_t^* in Section 2.1. At this juncture, further escalation of $Q_{C2,re,t}^*$ does not enhance $S_{2,re,t}^*$. When b_t attains its peak value of 1.05, the corresponding $Q_{C2,re,t}^*$ denotes the maximum installable reactive power compensation capacity on the low-voltage side. At this point, both b_t and K_t^* reach their limits, and $S_{2,re,t}^*$ approaches the OLTC's boundary, rendering additional reactive compensation futile.

2.3. The Influence of Electrical Distance from DG to Low-Voltage Side of OLTC on Reverse Power Flow Delivery Capacity

The electrical distance from the DG to the low-voltage side of the OLTC refers to the impedance relationship between these two points. A greater electrical distance Z_{DG}^* implies a larger voltage difference between these points when transmitting power flow, thus reducing the capacity for reverse power flow. The relationship between reverse power flow transmission capacity and the electrical distance from the DG to the low-voltage side of the OLTC can be derived as follows.

$$\begin{aligned}
 U_{3,t}^* - U_{2,t}^* &= \frac{P_{2,re,t}^* R_{DG}^* + Q_{2,re,t}^* X_{DG}^*}{U_{2,t}^*} \\
 U_{3,t}^* - U_{2,t}^* &= \frac{S_{2,re,t}^* Z_{DG}^* (\cos \varphi_T \sin \varphi_{Z_{DG}} + \sin \varphi_T \cos \varphi_{Z_{DG}})}{U_{2,t}^*} \\
 U_{3,t}^* - U_{2,t}^* &= \frac{S_{2,re,t}^* Z_{DG}^* c_{DG}}{U_{2,t}^*}
 \end{aligned} \tag{20}$$

$$U_{3,t}^* = \frac{U_{3,t}}{U_{1N}/K_t} = \frac{1.05U_{2N} \cdot K_t^* \cdot K_N}{U_{1N}} = \frac{1.05 \cdot K_t^*}{1.1} \quad (21)$$

$$S_{2,re,t}^* = \frac{(1.05 - b_t) b_t \cdot (K_t^*)^2}{1.21 \cdot Z_{DG}^* c_{DG}} \quad (22)$$

The differential $dU_{32,t}^*$ refers to the deviation between the voltage of the DG node and the per unit voltage at the low-voltage side of the OLTC. Parameters R_{DG}^* and X_{DG}^* denote the resistance and reactance from the DG to the OLTC's low-voltage side, respectively. The power factor of the impedance $\cos\varphi_{Z_{DG}}$ correlates to the DG towards the OLTC's low-voltage side impedance. The coefficient c_{DG} is constant, while $U_{3,t}^*$ represents the per unit equivalent of the DG node voltage. The aim is to ascertain the maximum achievable $S_{2,re,t}^*$ under the influence of Z_{DG}^* , leading K_t^* and $U_{3,t}^*$ to be maximized. The maximum $U_{3,t}^*$ is detailed in Equation (21). Substituting Equation (21) into Equation (20) yields the relationship between $S_{2,re,t}^*$ and Z_{DG}^* , as expressed in the equation. It shows an inverse relationship, illustrated in Figure 5, between $S_{2,re,t}^*$ and Z_{DG}^* .

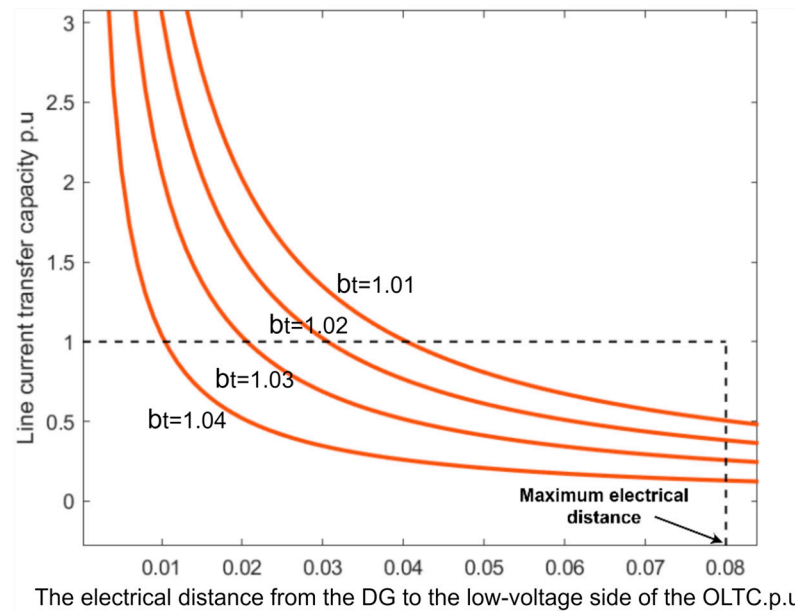


Figure 5. The relationship between electrical distance and the magnitude of reverse flow delivery.

The horizontal axis represents Z_{DG}^* , while the vertical axis denotes $S_{2,re,t}^*$. In the plot, a series of inverse proportional curves showcase the variation of $S_{2,re,t}^*$ with Z_{DG}^* as b_t decreases. As Z_{DG}^* increases, $S_{2,re,t}^*$ diminishes. Moreover, at a constant Z_{DG}^* , larger b_t values result in smaller $S_{2,re,t}^*$. This occurs because the maximum Z_{DG}^* remains constant, hence as b_t increases, the $dU_{32,t}^*$ decreases, thereby reducing $S_{2,re,t}^*$.

3. Actual Maximum Transfer Capacity of Reverse Power Flow in the Distribution Network

Sections 2.1–2.3 analyzed the impacts of the OLTC transformation ratio K_t^* , the reactive power compensation capacity on the low-voltage side of the OLTC $Q_{C2,re,t}^*$, the electrical distance from the DG to the low-voltage side of the OLTC Z_{DG}^* , and the OLTC's low-voltage side voltage parameter b_t on $S_{2,re,t}^*$. Section 2.1 states that $S_{2,re,t}^*$ increases with larger K_t^* and b_t , reaching up to approximately 84% of the OLTC's rated capacity when both are at maximum values. Section 2.2 explains that increasing K_t^* and b_t to enhance $S_{2,re,t}^*$ requires reactive support from $Q_{C2,re,t}^*$ on the low-voltage side, necessitating additional reactive equipment, which raises cost. However, given sufficient reactive power,

$S_{2,re,t}^*$ achieves the maximum value from Section 2.1. Section 2.3 notes that $S_{2,re,t}^*$ decreases as Z_{DG}^* rises and points out a contradiction with prior conclusions when Z_{DG}^* is fixed and a smaller b_t results in a larger $S_{2,re,t}^*$.

In the above scenario, analyzing from the perspective of the OLTC, an increase in b_t results in a larger $S_{2,re,t}^*$. However, when analyzed from the perspective of the line to the low-voltage side of the DG to OLTC, an increase in b_t results in a smaller $S_{2,re,t}^*$. Therefore, it is necessary to select an appropriate b_t to maximize $S_{2,re,t}^*$ after comprehensively considering various influencing factors. The actual maximum reverse power transfer capacity of distribution networks, $S_{2,re,t,max}^*$, can be expressed as:

$$S_{2,re,t,max}^* = \min \{(7), (9), (22)\} \quad (23)$$

$$(7) = (9) = (22) \quad (24)$$

Equation (23) elucidates that the value of $S_{2,re,t,max}^*$ is determined as the minimum under three distinct constraints. Equation (24) indicates that $S_{2,re,t}^*$ should remain equal across the same time period under three different constraints, as they represent the same power flow. Figure 6 demonstrates the relationship among $S_{2,re,t}^*$, $Q_{C2,re,t}^*$, Z_{DG}^* , and b_t when K_t^* is set to K_{max}^* .

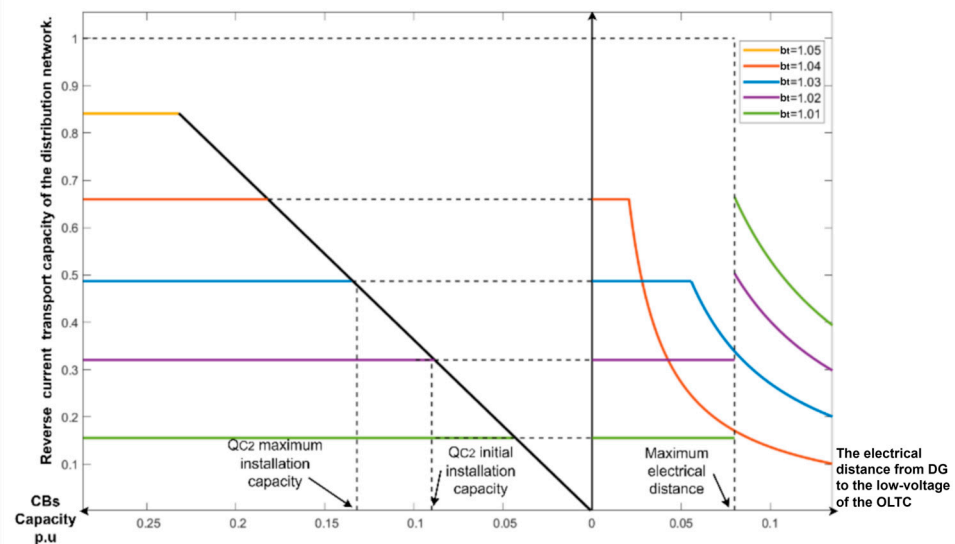


Figure 6. Real power reverse flow transmission capacity distribution map of the distribution network.

The horizontal axis of the graph has its positive direction labeled as Z_{DG}^* , and its negative direction is labeled as $Q_{C2,re,t}^*$. The vertical axis is $S_{2,re,t}^*$. On the right side of the vertical axis, inverse proportion curves in different colors represent the variation of $S_{2,re,t}^*$ with Z_{DG}^* under various b_t conditions. On the left side, a black linear function indicates the relationship of $S_{2,re,t}^*$ with $Q_{C2,re,t}^*$ under different b_t conditions. The colored horizontal lines on both sides of the vertical axis signify the value of $S_{2,re,t}^*$ when $K_t^* = K_{max}^*$, across different b_t scenarios. When $K_t^* = K_{max}^*$ and reactive power compensation $Q_{C2,re,t}^*$ is sufficient, without considering Z_{DG}^* constraints, $S_{2,re,t,max}^*$ can reach approximately 84% of the OLTC's rated capacity. However, in practical operation, with Z_{DG}^* limitations considered, $S_{2,re,t,max}^*$ is limited to about 50% of the OLTC's rated capacity. Refer to Figure 1 for an equivalent circuit diagram, taking the high-voltage side node of the OLTC as the balance node. The specific calculation process of $S_{2,re,t,max}^*$ is illustrated in Table 1.

Table 1. The specific computational procedure for $S_{2,re,t,max}^*$.

Step	Methodology
①	Input the data for the distribution network, including the internal impedance of the OLTC Z_T^* , the maximum transformation ratio K_{max}^* , the rated capacity S_N^* , the initial reactive power compensation on the low-voltage side, and the impedance from the equivalent DG node to the OLTC's low-voltage side Z_{DG}^* .
②	Input the initial value of $S_{2,re,t}^*$, designating this initial value as the rated capacity of the OLTC, S_N^* . Set the transformation ratio for the OLTC as K_{max}^* .
③	Substitute the input data from Step 2 into Equation (7) to calculate the value of b_t .
④	Evaluate whether b_t falls within the permissible range. If it does not, revert to Step 1 and decrease the initial value of $S_{2,re,t}^*$ accordingly until b_t remains within the allowable limits. The current value of $S_{2,re,t}^*$ represents the maximum transmission capacity $S_{2,re,max}^*$ of the distribution network, considering only the OLTC transformation ratio.
⑤	Input the values of b_t , K_{max}^* , and $S_{2,re,t}^*$ into Equations (4) and (20) to calculate the magnitude of $U_{3,t}^*$.
⑥	Determine if $U_{3,t}^*$ exceeds the limit. If it does, decrease $S_{2,re,t}^*$ and return to Step 2. $U_{3,t}^*$ is the highest voltage point in the distribution network. If $U_{3,t}^*$ is within limits, other nodes will also be within limits.
⑦	Substitute the calculated $S_{2,re,t}^*$ from Step 5 into Equation (19) to determine the required reactive compensation capacity $Q_{C2,re,t}^*$.
⑧	To obtain $S_{2,re,t}^*$ represents the maximum transmission capacity $S_{2,re,max}^*$ of the distribution network under reverse flow, taking all factors into account, as well as the required reactive power compensation capacity, producing the output results.

The specific flowchart is shown in Figure 7. The aforementioned steps calculate the maximum reverse power flow transfer capacity for this distribution network. When the required reverse power flow exceeds this value, the DG output needs reduction. Conversely, if the reverse power flow is less than this value, it can be transferred smoothly. Additionally, the input K_t^* in Step 2 can be reduced to optimize the operational cost of the distribution network.

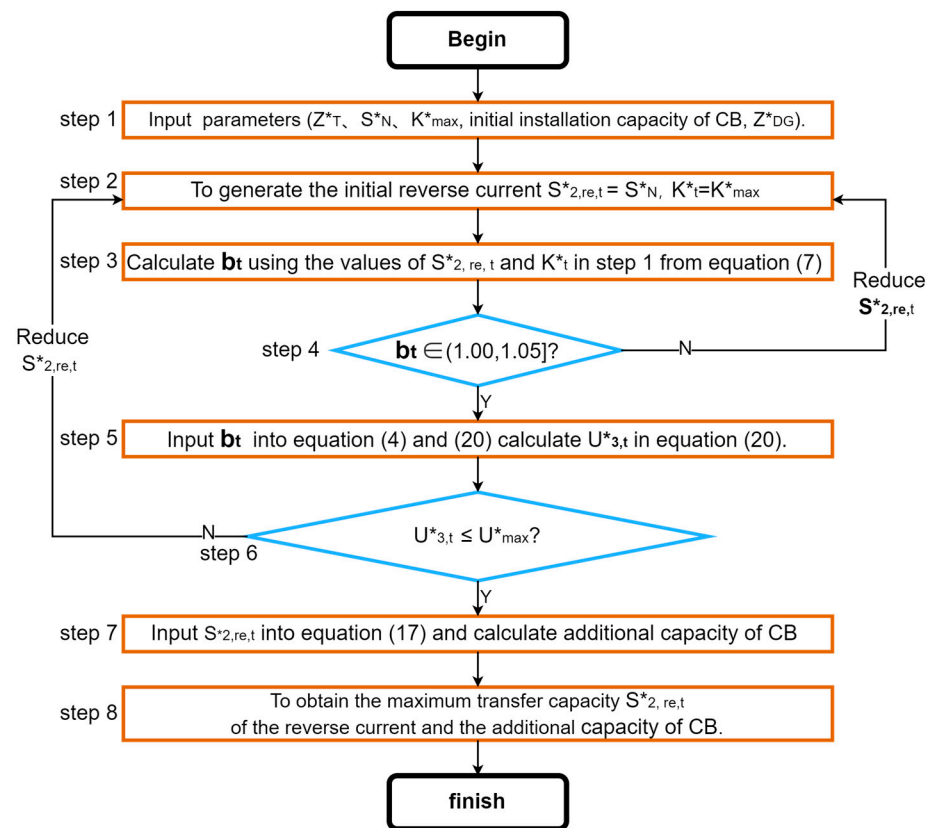


Figure 7. Calculation procedure for maximum transmission capacity of reverse power flow in distribution networks.

4. Joint Operation Optimization Model for Multi-Distribution Networks with SOP Considering Reverse Power Flow

Since the maximum value of $S_{2, re, t, max}^*$ in the distribution network is only about 50% of the OLTC's rated capacity, it limits further development. To address this issue, this paper employs a soft open point (SOP) to flexibly interconnect multiple distribution grids. This enables sharing of backward power transfer capacities among sub-distribution networks and enhances the grid's capacity to handle reverse flow under high penetration of the DG. Figure 8 illustrates the SOP interconnection schematic, where the SOP interlinks two distribution networks at an equivalent power source node. When the reverse power flow in distribution network 1 exceeds its maximum transferable capacity, the SOP diverts the excess power to distribution network 2, which then transmits it to the upstream grid, thus reducing occurrences of wind and solar energy curtailment.

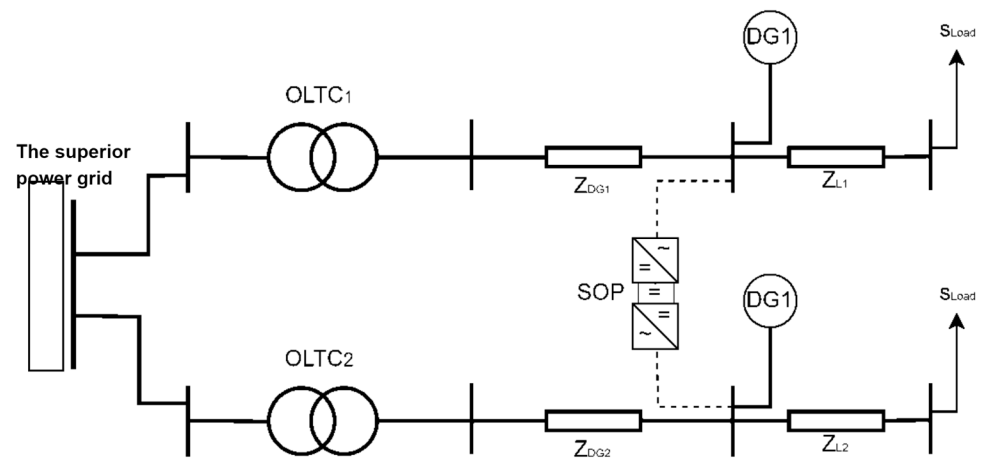


Figure 8. The SOP interconnects two distribution networks.

4.1. Fundamental Principles and Mathematical Models of SOP

This paper employs a back-to-back voltage source converter (B2B VSC) to implement the SOP. The number of ports is determined by the count of AC/DC bidirectional converters, which are internally connected through a DC/DC converter. The AC side of the AC/DC converter links to the distribution network feeders, while the DC side connects to the DC/DC converter, enabling interconnection between different feeders. The basic topology is illustrated in the Figure 9.

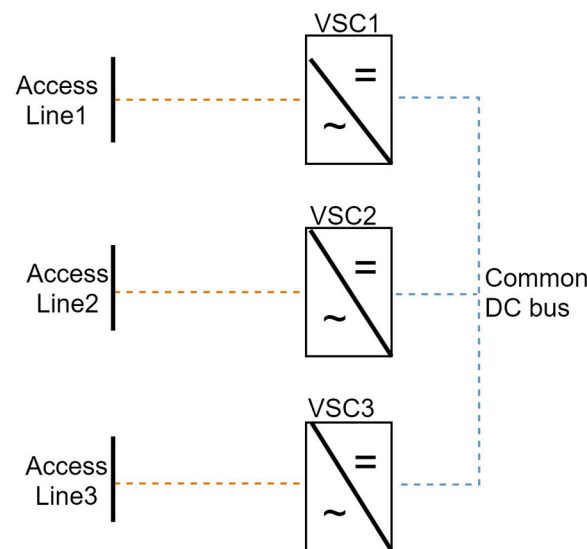


Figure 9. The SOP's basic topological structure.

Under normal operating conditions, the two-end SOP selects a PQ-Vdc control mode, while the three-end SOP employs a PQ-PQ-Vdc Q control mode. This means that one VSC regulates the DC-side voltage, and the other VSCs control the power being transmitted. The operational variables of the SOP include the active and reactive power transmitted by the VSCs, with specific operational constraints as follows [28,29]:

$$\sum_{i \in \Omega_{SOP}} (P_{i,t}^{AC} - P_{i,t,loss}^{AC}) = 0 \quad (25)$$

$$S_i^{AC} \geq \sqrt{(P_{i,t}^{AC})^2 + (Q_{i,t}^{AC})^2} \quad (26)$$

$$S_i^{AC} \geq P_{i,t}^{AC} \geq -S_i^{AC} \quad (27)$$

$$S_i^{AC} \geq Q_{i,t}^{AC} \geq -S_i^{AC} \quad (28)$$

The set Ω_{SOP} contains the port nodes of the SOP. The terms $P_{i,t}^{AC}$ and $Q_{i,t}^{AC}$ represent the active and reactive power injected by port i of the SOP at time period t . The terms P_t^{DC} and $P_{t,loss}^{SOP}$ are the active power injected by the SOP's DC/DC converter and the total active power loss at time period t , respectively. The variable S_i^{AC} denotes the capacity of the AC/DC converter at port i of the SOP, and S^{DC} represents the capacity of the SOP's DC/DC converter. When the SOP transmits a significant amount of power, its internal power loss becomes non-negligible. Given that multiple VSCs are identical in model and symmetrically positioned, the analysis focuses on one VSC_i , and its equivalent model is illustrated in Figure 10.

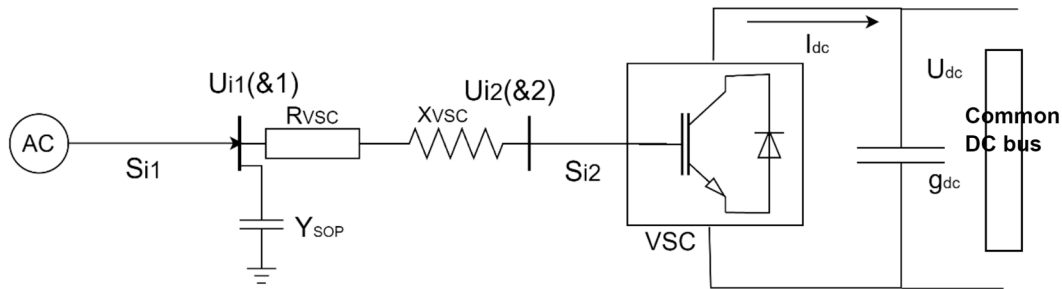


Figure 10. Diagram of VSC AC/DC model, considering impedance.

In the Figure 10, S_{i1} represents the SOP bus's injected SOP power, while S_{i2} denotes the apparent power injected by the VSC at the converter bus. $U_{i1} \angle \delta_1$ and $U_{i2} \angle \delta_2$ correspond to the voltage and phase angle at the SOP bus and VSC bus, respectively. The phase angle difference $\Delta \delta$ indicates the phase angle of VSC pulse modulation, determining the magnitude and direction of SOP injected power. U_{dc} and I_{dc} stand for the DC-side current and voltage of the VSC, and g_{DC} represents the equivalent conductance on the VSC's DC side. R_{VSC} and X_{VSC} are the equivalent resistance and reactance of the VSC. Specific constraints include:

- DC-side KCL balance constraint:

$$I_{dc,t} = g_{DC} \cdot U_{dc,t} \quad (29)$$

- AC/DC voltage pulse modulation constraints:

$$U_{i2,t} = \frac{\mu M_i}{\sqrt{2}} U_{dc,t} \quad (30)$$

In this context, μ represents the direct current voltage utilization rate, while M_i denotes the modulation index of the pulse signal.

- Alternating current measurement involves balancing constraints of active and reactive power:

$$P_{i,t}^{AC} = P_{i1,t} = \frac{U_{i1,t}^2}{R_{VSC}} - U_{i1,t}U_{i2,t}U_{dc,t}(G_i \sin \Delta \delta + B_i \cos \Delta \delta) \quad (31)$$

$$Q_{i,t}^{AC} = Q_{i1,t} = -\frac{U_{i1,t}^2}{X_{VSC}} + U_{i1,t}U_{i2,t}U_{dc,t}(B_i \cos \Delta \delta - G_i \sin \Delta \delta) \quad (32)$$

where G_i and B_i represent the AC filter admittance of the SOP, denoted as Y_{SOP} .

- Power loss constraint of VSC:

$$P_{i,t,loss}^{AC} = \frac{U_{i1,t}^2}{R_{VSC}} - U_{i1,t}U_{i2,t}U_{dc,t}(G_i \sin \Delta \delta + B_i \cos \Delta \delta) - \frac{(\mu M_i)^2}{2R_{VSC}}U_{dc} - I_{dc} \cdot U_{dc} \quad (33)$$

The active power loss of VSC equates to the power injected on the AC side minus the power injected on the DC side. In the context of SOP involvement in distribution network optimization, there are five unknowns in the five constraints stated above. When employing PQ control, the unknowns are: U_{dc} , I_{dc} , $P_{loss,i}$, $\Delta \delta$, and M_i . Under QU control, the unknowns are: P_{i1} , I_{dc} , $P_{loss,i}$, $\Delta \delta$, and M_i , which can be solved based on optimization results.

4.2. Objective Function

The objective function aims to minimize the sum of the daily comprehensive operating costs of multiple distribution networks after joint operation. The daily comprehensive operating costs include: network loss cost $\mu_{loss}P_{m,t}^{loss}$, wind and solar curtailment cost $\mu_{DG}P_{DG,m,t}^{loss}$, and device (SOP, CB, OLTC) operation cost $C_{OP,m,t}$, as expressed in the following equation:

$$\min F_m = \sum_{m=1}^{\Omega} \sum_{t=1}^T (\mu_{loss}P_{m,t}^{*loss} + \mu_{DG}P_{DG,m,t}^{*loss} + C_{OP,m,t}) \quad (34)$$

$$P_{m,t}^{*loss} = \sum_{ij \in \Omega_B} I_{m,ij,t}^{*2} r_{m,ij}^* \quad (35)$$

$$P_{m,DG,t}^{*loss} = \sum_{i=1}^N (P_{m,i,t}^{*DG} + P_{m,i,t}^{*SOP} - P_{m,i,t}^{*L}) - P_{m,t}^{*loss} - \cos \varphi_m S_{m,re,max}^* \quad (36)$$

$$C_{m,OP,t} = C_{m,OP,t}^{SOP} + C_{m,OP,t}^{OLTC} + C_{m,OP,t}^{CB} \quad (37)$$

In this context, $P_{m,t}^{*loss}$ and $P_{m,DG,t}^{*loss}$ denote the per unit values of active network losses and curtailed wind and solar resources, respectively, for distribution network m during period t . The coefficients μ_{loss} and μ_{DG} represent the cost weights for network losses and renewable energy curtailment. $P_{m,DG,i,t}^*$ and $P_{m,load,i,t}^*$ denote the dispatched power from distributed generation and load demand at node i in distribution network m during period t . Costs $C_{OP,m,t}^{SOP}$, $C_{OP,m,t}^{OLTC}$, and $C_{OP,m,t}^{CB}$ correspond to the operational actions of the SOP, OLTC, and CB, respectively.

4.3. Constraints

4.3.1. Power Distribution Grid Flow Constraint

To model the distribution network using the *Distflow* branch model [18], the power flow constraints of distribution network m can be expressed as:

$$\sum_{ji \in \Omega_b} (P_{m,ji,t}^* - I_{m,ji,t}^{*2} r_{m,ji}^*) + P_{m,i,t}^* = \sum_{ik \in \Omega_b} (P_{m,ik,t}^*) \quad (38)$$

$$\sum_{ji \in \Omega_b} (Q_{m,ji,t}^* - I_{m,ji,t}^{*2} x_{ji}^*) + Q_{m,i,t}^* = \sum_{ik \in \Omega_b} (Q_{m,ik,t}^*) \quad (39)$$

$$U_{m,i,t}^{*2} - U_{m,j,t}^{*2} - 2(P_{m,ij,t}^* r_{m,ij}^* + Q_{m,ij,t}^* x_{m,ij}^*) + (r_{m,ij}^{*2} + x_{m,ij}^{*2}) I_{m,ij,t}^{*2} = 0 \quad (40)$$

$$I_{m,ij,t}^{*2} U_{m,i,t}^{*2} = P_{m,ij,t}^{*2} + Q_{m,ij,t}^{*2} \quad (41)$$

$$P_{m,i,t}^* = P_{m,i,t}^{*DG} + P_{m,i,t}^{*SOP} - P_{m,i,t}^{*L} \quad (42)$$

$$Q_{m,i,t}^* = Q_{m,i,t}^{*DG} + Q_{m,i,t}^{*SOP} + Q_{m,i,t}^{*CB} - Q_{m,i,t}^{*L} \quad (43)$$

$$U_{min}^{*2} \leq U_{m,i,t}^{*2} \leq U_{max}^{*2} \quad (44)$$

$$I_{m,ij,t}^{*2} \leq I_{max}^{*2} \quad (45)$$

In the distribution network m , $P_{m,ji,t}^*$ and $Q_{m,ji,t}^*$ denote the active and reactive power injected at node i , respectively, while $P_{m,ik,t}^*$ and $Q_{m,ik,t}^*$ represent the active and reactive power flowing out from node i . The resistance and reactance on line ij are $r_{m,ji}$ and $x_{m,ji}$, respectively. The active power injection at node i during time period t for the DG, SOP, and load in distribution network m are $P_{m,DG,i,t}^*$, $P_{m,SOP,i,t}^*$, and $P_{m,Load,i,t}^*$, respectively. The reactive power injections at node i during time period t for the DG, SOP, CBs, and load are $Q_{m,DG,i,t}^*$, $Q_{m,CB,i,t}^*$, $Q_{m,SOP,i,t}^*$, and $Q_{m,Load,i,t}^*$, respectively. The node voltage at i is $U_{i,t}^*$, with U_{min}^* and U_{max}^* as the minimum and maximum voltage values, respectively. The branch current ij and the maximum allowable current on the line are $I_{m,ij,t}^*$ and I_{max}^* , respectively.

4.3.2. Distributed Generation Output Constraint

$$0 \leq P_{m,DG,i,t} \leq P_{m,DG,i,t}^{max} \quad (46)$$

$$Q_{m,DG,i,t} = P_{m,DG,i,t} \tan \varphi_{m,DG} \quad (47)$$

In which $P_{DG,i,t}$ and $P_{DG,i,t}^{max}$ represent the active power output and maximum active power output at node i during time period t , respectively. Meanwhile, $Q_{DG,i,t}$ and $Q_{DG,max,i}$ denote reactive power output and maximum reactive power output. φ_{DG} is the power factor angle of the DG.

4.3.3. Reactive Power Compensation Constraint

$$Q_{m,CB,i,t} = N_{m,CB,i,t} \cdot q_{m,CBs,i} \quad (48)$$

$$\sum_{t=1}^{24} |N_{m,CBs,i,t} - N_{m,CB,i,t-1}| \leq \Delta CB_{m,i,max} \quad (49)$$

$$0 \leq N_{m,CB,i,t} \leq N_{m,i,max} \quad (50)$$

In the distribution network m , $q_{m,CB,i}$ denotes the capacitance at node i for a single capacitor group (and reactor for the high-voltage side). $N_{m,CB,i,t}$ represents the number of engaged CB groups. $\Delta CB_{m,i,max}$ signifies the maximum switching operations constraint for CB stages within the scheduling period. $N_{m,i,max}$ refers to the maximum number of groups that CB can employ.

4.3.4. OLTC Operating Constraints

$$U_{m,i,t} = U_{m,j,t} k_{m,ij,t} \quad (51)$$

$$k_{m,ij,t} = k_{m,ij,0} + K_{m,ij,t} \Delta k_{m,ij} \quad (52)$$

$$\sum_{t=1}^{24} |K_{m,ij,t} - K_{m,ij,t-1}| \leq \Delta K_{m,ij,max} \quad (53)$$

$$K_{m,ij,min} \leq K_{m,ij,t} \leq K_{m,ij,max} \quad (54)$$

In the distribution network m , $k_{m,ij,t}$ represents the transformation ratio of the OLTC between nodes i and j during time period t . $k_{ij,0}$ denotes the original transformation ratio. $K_{m,ij,t}$ signifies the tap position of the OLTC during time period t . $K_{ij,min}$ and $K_{m,ij,max}$ are the minimum and maximum tap positions, respectively. $\Delta K_{m,ij,max}$ constrains the maximum number of tap operations of the OLTC within one scheduling cycle.

4.3.5. SOP Constraints

Equations (26)–(34).

$$C_{m,OP,t}^{SOP} = P_{i,t,loss}^{AC} \quad (55)$$

4.3.6. Reverse Power Flow Transmission Capacity Constraint in Distribution Networks

Equations (23) and (24).

4.4. Model Transformation and Solution Process

In the model mentioned above, the variables involve both SOP real and reactive power outputs as continuous variables, and actions of the OLTC and CB components as discrete variables. Essentially, it is a mixed-integer nonlinear programming problem, difficult to solve directly for optimization. This paper employs second-order cone relaxation to transform it into a mixed-integer second-order cone programming model, thus allowing the direct use of the CPLEX commercial solver for resolution. Nonlinear elements such as $I_{m,ji,t}^2$ and $U_{m,i,t}^2$ in Equations (41)–(46) are replaced by $l_{m,ji,t}$ and $V_{m,i,t}$. However, Equation (42) remains nonlinear, so it is relaxed into the following second-order cone constraints:

$$\left\| [2P_{m,ij,t} 2Q_{m,ij,t} l_{m,ji,t} - V_{m,i,t}]^T \right\|_2 < l_{m,ji,t} + V_{m,i,t} \quad (56)$$

Constraints (27) containing quadratic nonlinearity can be transformed into the following rotated second-order cone constraints.

$$(S_i^{AC})^2 \geq (P_{i,t}^{AC})^2 + (Q_{i,t}^{AC})^2 \quad (57)$$

The OLTC-related constraint (53) will transform after introducing $V_{i,t}$:

$$V_{m,i,t} = k_{m,ij,t}^2 V_{m,j,t} \quad (58)$$

At this point, one needs to handle the quadratic terms, introducing the binary variable $b_{m,ij,t,k}$ to represent them in the following form.

$$K_{m,ij,t} = \sum_{k=0}^{2K_{m,ij,max}} (k - K_{m,ij,max}) b_{m,ij,t,k} \quad (59)$$

$$\sum_{k=0}^{2K_{m,ij,max}} b_{m,ij,t,k} = 1, b_{m,ij,t,k} \in \{0,1\} \quad (60)$$

After substitution, the result is:

$$V_{m,i,t} = \sum_{k=0}^{2K_{m,ij,max}} (k_{m,ij,0} + (k - K_{m,ij,max})\Delta k_{ij})^2 V_{m,j,t} b_{m,ij,t,k} \quad (61)$$

At this point, the product $V_{j,t} b_{ij,t,k}$ remains nonlinear and can be expressed by the variable $V_{ij,t,k}^b$. Accordingly, Equation (53) can be represented as:

$$V_{m,i,t} = \sum_{k=0}^{2K_{ij,max}} (k_{m,ij,0} + (k - K_{m,ij,max})\Delta k_{m,ij})^2 V_{m,ij,t,k}^b \quad (62)$$

Constraint (45) is expressed as:

$$U_{i,min}^2 b_{m,ij,t,k} < V_{m,ij,t,k}^b < U_{i,max}^2 b_{m,ij,t,k} \quad (63)$$

$$U_{i,min}^2 (1 - b_{m,ij,t,k}) < V_{m,i,t} - V_{m,ij,t,k}^b < U_{i,max}^2 (1 - b_{m,ij,t,k}) \quad (64)$$

For constraints (54) and (55), auxiliary variables $K_{m,ij,t}^+$ and $K_{m,ij,t}^-$ could be introduced, representing the number of tap positions for the OLTC adjusted upwards or downwards, respectively. Thus, constraints (54) and (55) can be expressed as:

$$\sum_{t=1}^{24} |K_{m,ij,t}^+ + K_{m,ij,t}^-| \leq \Delta K_{ij,max} \quad (65)$$

$$K_{m,ij,t} - K_{m,ij,t-1} = K_{m,ij,t}^+ - K_{m,ij,t}^- \quad (66)$$

$$K_{m,ij,t}^+ \geq 0, K_{m,ij,t}^- \geq 0 \quad (67)$$

Similarly, the constraints regarding CB can also be expressed as:

$$\sum_{t=1}^{24} |N_{m,i,t}^+ + N_{m,i,t}^-| \leq \Delta N_{m,i,max} \quad (68)$$

$$N_{m,i,t} - N_{m,i,t-1} = N_{m,i,t}^+ - N_{m,i,t}^- \quad (69)$$

$$N_{m,i,t}^+ \geq 0, N_{m,i,t}^- \geq 0 \quad (70)$$

The objective function's $C_{m,OP,t}$ can be expressed as:

$$C_{m,OP,t} = P_{i,t,loss}^{AC} + \sum_{ij \in \Omega_o} C_{OLTC} (K_{m,ij,t}^+ + K_{m,ij,t}^-) + \sum_{i \in \Omega_{CB}} C_{CB} (N_{m,i,t}^+ + N_{m,i,t}^-) \quad (71)$$

In this context, Ω_o and Ω_{CB} denote the set of branches with the OLTC and the set of nodes with CB in distribution network m , respectively.

5. Case Study

5.1. Case Study Parameters

This paper uses three improved IEEE 33 distribution networks to conduct a case study analysis on optimization strategies for multi-distribution network joint operation with the SOP under reverse power flow consideration. The topological structures of the three distribution networks are shown in Figure 11. The DG output and load forecast for each distribution network on a certain day are illustrated in Figure 12. It can be seen that

in distribution network A, the DG output is high, maintaining reverse power flow throughout the day. In distribution network B, the DG output is moderate, with reverse power flow appearing during early morning and midday. In distribution network C, the DG output is low, with reverse power flow occurring only during the early morning.

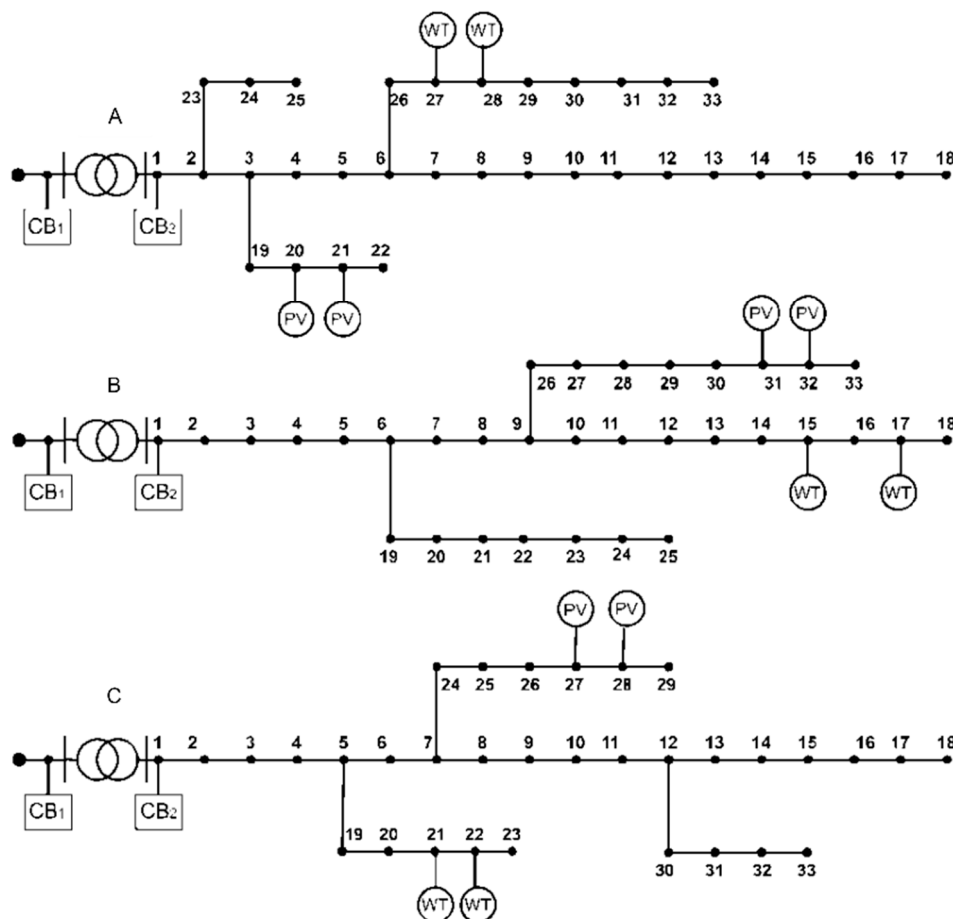
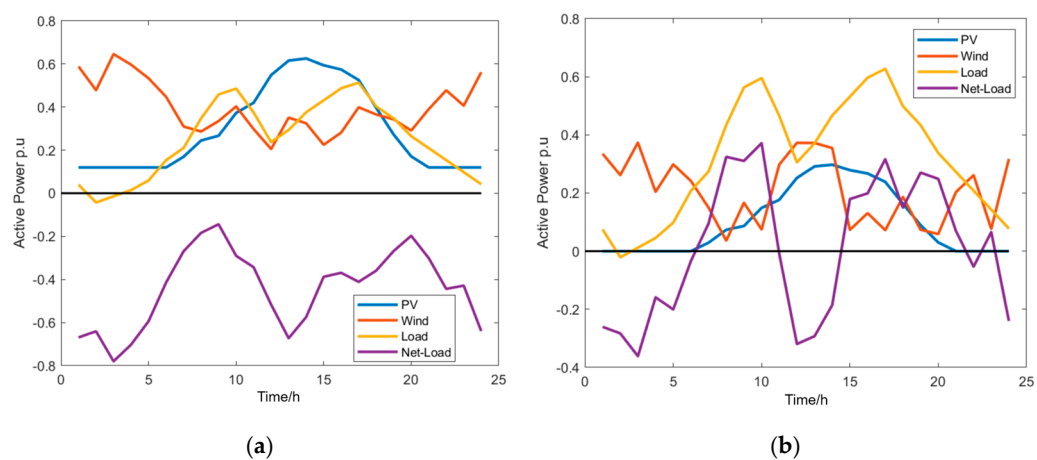


Figure 11. Topology of distribution networks A, B, and C.



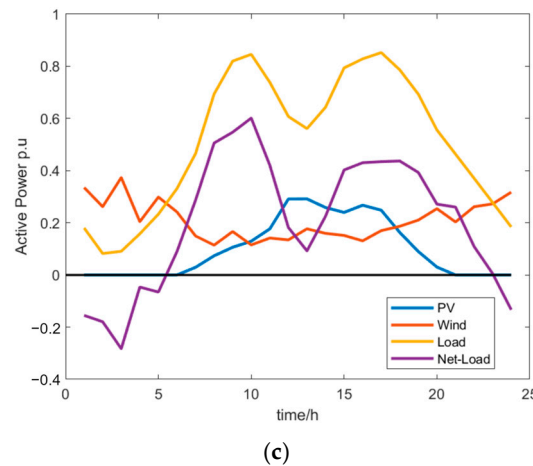


Figure 12. The load and DG output conditions of distribution networks A, B, and C, and (a–c) respectively represent distribution networks A, B, and C.

The OLTC parameters in the three distribution networks are identical, with a rated transformation ratio K_N of 110/11, a rated voltage U_N of 110 kV, a rated capacity S_N of 20 MVA, a rated power factor $\cos\varphi_T$ of 0.85, P_k of 112 kW, $U_k\%$ of 10.5%, high-voltage side taps of $\pm 1.25\% \times 8$, and a maximum number of actions within a scheduling cycle $K_{m,ij,max} = 10$.

In the reactive power compensation on the low-voltage side, the capacity of a single capacitor $q_{m,CB,i} = 1\text{MVA}$, with a total installation group number $N_{m,i,max} = 5$, and a maximum number of operations per dispatch period $\Delta N_{m,i,max} = 10$.

In the utilized SOP, each VSC has a capacity of 5.0 MVA with a loss coefficient of 0.02. For the distribution networks A, B, and C, nodes connected to the SOP are selected according to the equivalent source method: A.5, B.14, and C.6 respectively. The model was tested on a PC with a 2.2 GHz main frequency and 32GB of memory, using the MATLAB R2020b simulation platform. Embedded in this platform is the YALMIP mixed-integer optimization toolbox, which calls CPLEX version 12.8 for optimization solving [30].

5.2. Case Analysis

To verify that the reverse power flow transmission capacity is insufficient in the DG high-penetration distribution networks, and by operating multiple distribution networks jointly through the SOP, the reverse power flow transmission capacity can be increased to reduce wind and solar curtailment, thus improving the operational efficiency of distribution networks. The following optimization scheme is established:

Scenario 1: Distribution networks A, B, and C operate independently.

Scenario 2: Distribution networks A and B operate jointly through a dual-port SOP.

Scenario 3: Distribution networks A, B, and C operate jointly through a tri-port SOP.

Scenario 1 serves as the original scheme. It examines the distribution network under varying operational characteristics such as the DG output, load demand, and topology. It analyzes the actions of the OLTC and CB, as well as wind and solar curtailment and network losses during independent operation. This serves as a reference for subsequent Scenarios (2 and 3) which involve the combined operation of two distribution networks. However, they differ in operational characteristics to explore optimization effects after interconnection. Scenario 4 involves three distribution networks operating jointly, allowing the exploration of how the number of interconnected distribution networks impacts optimization results. Specifically, it assesses whether more interconnected networks lead to improved optimization outcomes.

5.2.1. Analysis of Optimization Results During Independent Operation of Distribution Networks

The scenario 1 analyzes the scenario where different distribution networks with varying operational characteristics rely solely on the OLTC and CB for optimization during independent operation. Figures 13–15 show the results for distribution networks A, B, and C.

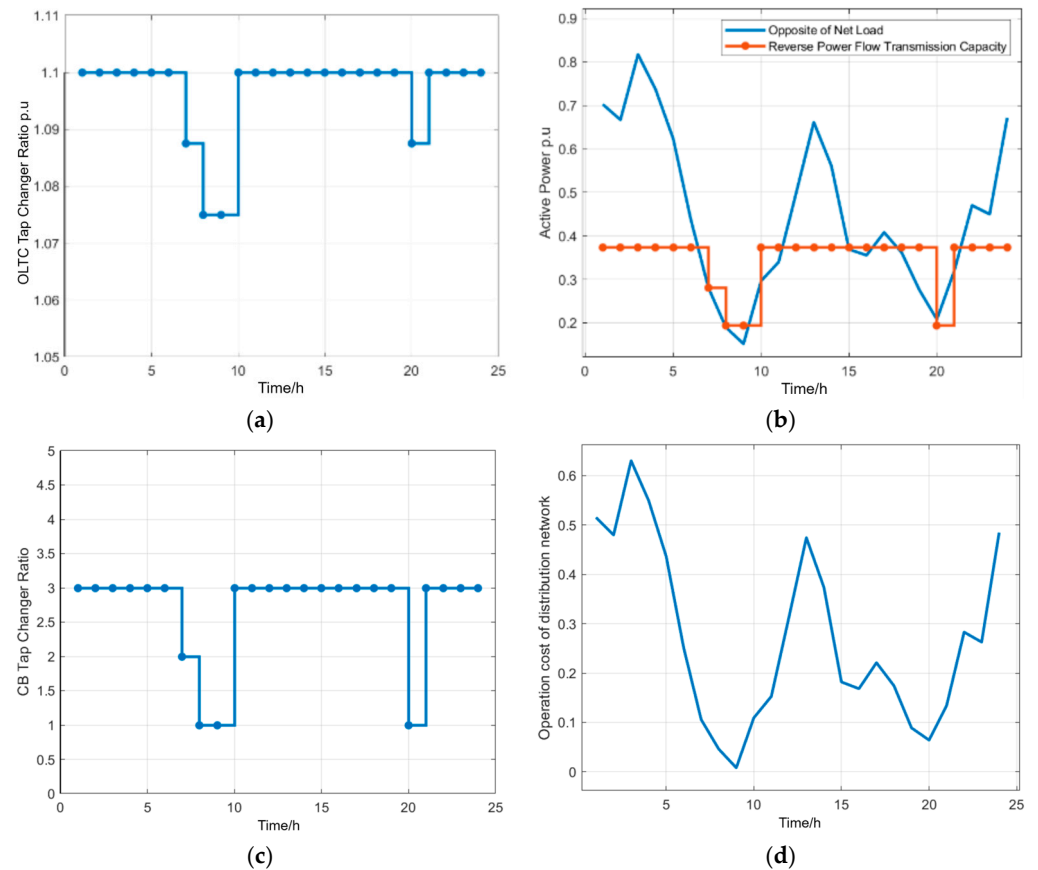
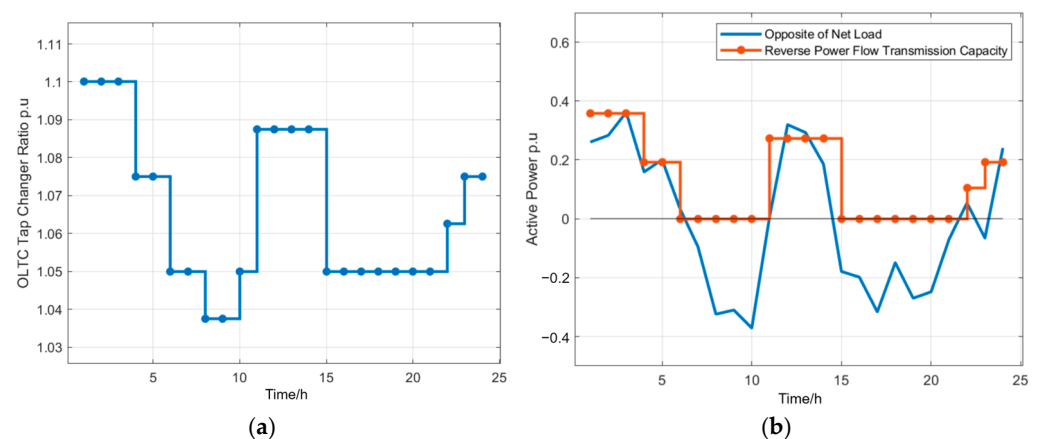


Figure 13. Scenario 1 presents the operational results of distribution network A: (a) shows the tap position changes of the OLTC; (b) illustrates the relationship between the negative net load and the reverse power flow capacity of the distribution network; (c) indicates the tap position changes of the CB; (d) represents the operational cost of the distribution network. In distribution network A, due to significant wind and solar curtailment, the operational cost curve aligns with the net load curve.



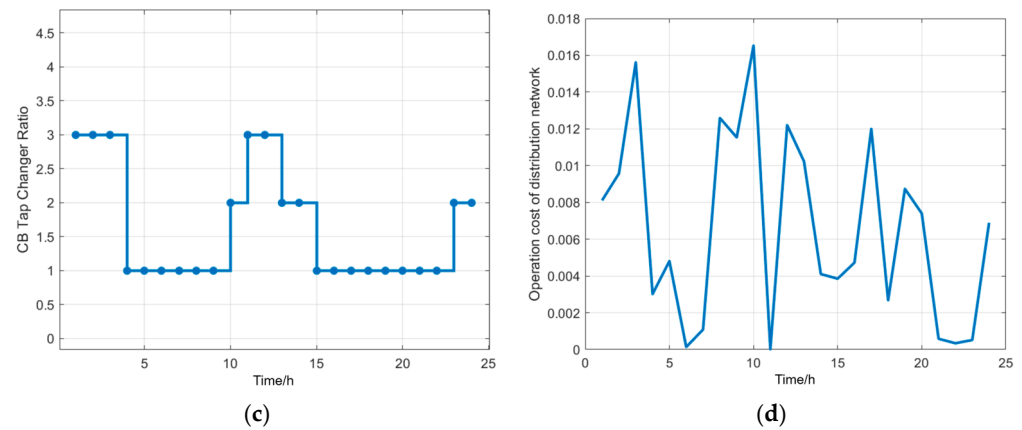


Figure 14. Scenario 1 presents the operational results of distribution network B, (a) as the operational status of the OLTC, (b) as the curtailment of wind and solar, (c) as the CB action status, and (d) as operational costs. The phenomenon of wind and solar curtailment is less prominent in distribution network B due to smaller reverse power flows. Consequently, operational costs primarily depend on network losses. When reverse power flow is minimal, OLTC tap positions can be lowered. Although this may increase the operation costs of the OLTC and CB, it elevates voltage in subsequent lines, thereby reducing overall active power losses.

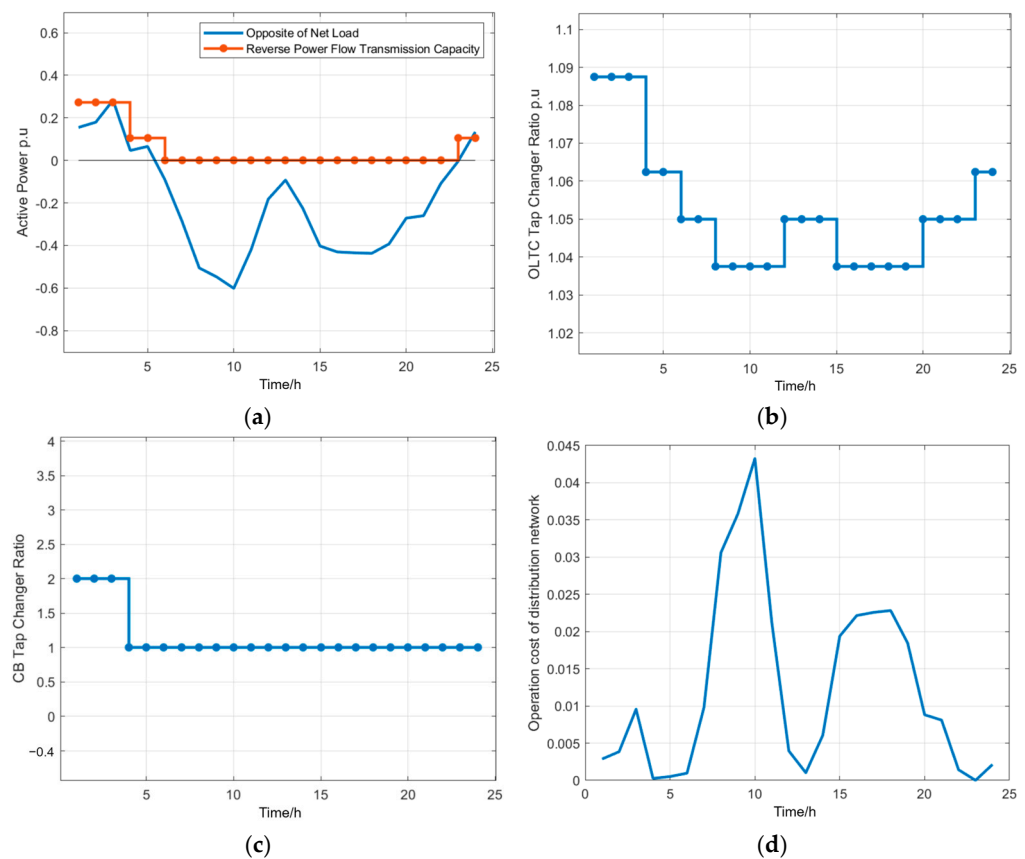


Figure 15. Scenario 1 presents the operational results of distribution network C, (a) as the operational status of the OLTC, (b) as the curtailment of wind and solar, (c) as the CB action status, and (d) as operational costs. In distribution network C, due to the minimal reverse power flow, there is almost no curtailment of wind or solar energy.

Comparing the operation of distribution networks A, B, and C, one can observe distinct DG placements. While network A's DG lies closest to the OLTC and network B's DG is further, all three networks show a similar active per unit value of maximal reverse power flow transfer capacity, approximately 0.37. Furthermore, permissible reverse

power flow capacity varies directly with the tap changer ratio; a larger ratio increases capacity yet demands more reactive power compensation. Even at maximum tap ratio, reverse flow capacity remains below 50% of the OLTC's rated capacity, significantly lower than the 80% target set in the design phase.

5.2.2. Optimized Results of the Joint Operation of Distribution Networks A and B

In scenario 2, the situation of distribution network A and B after joint operation by SOP is analyzed. It can be seen in Figure 16 that the operation of distribution network A after joint operation has been improved compared with that after separate operation.

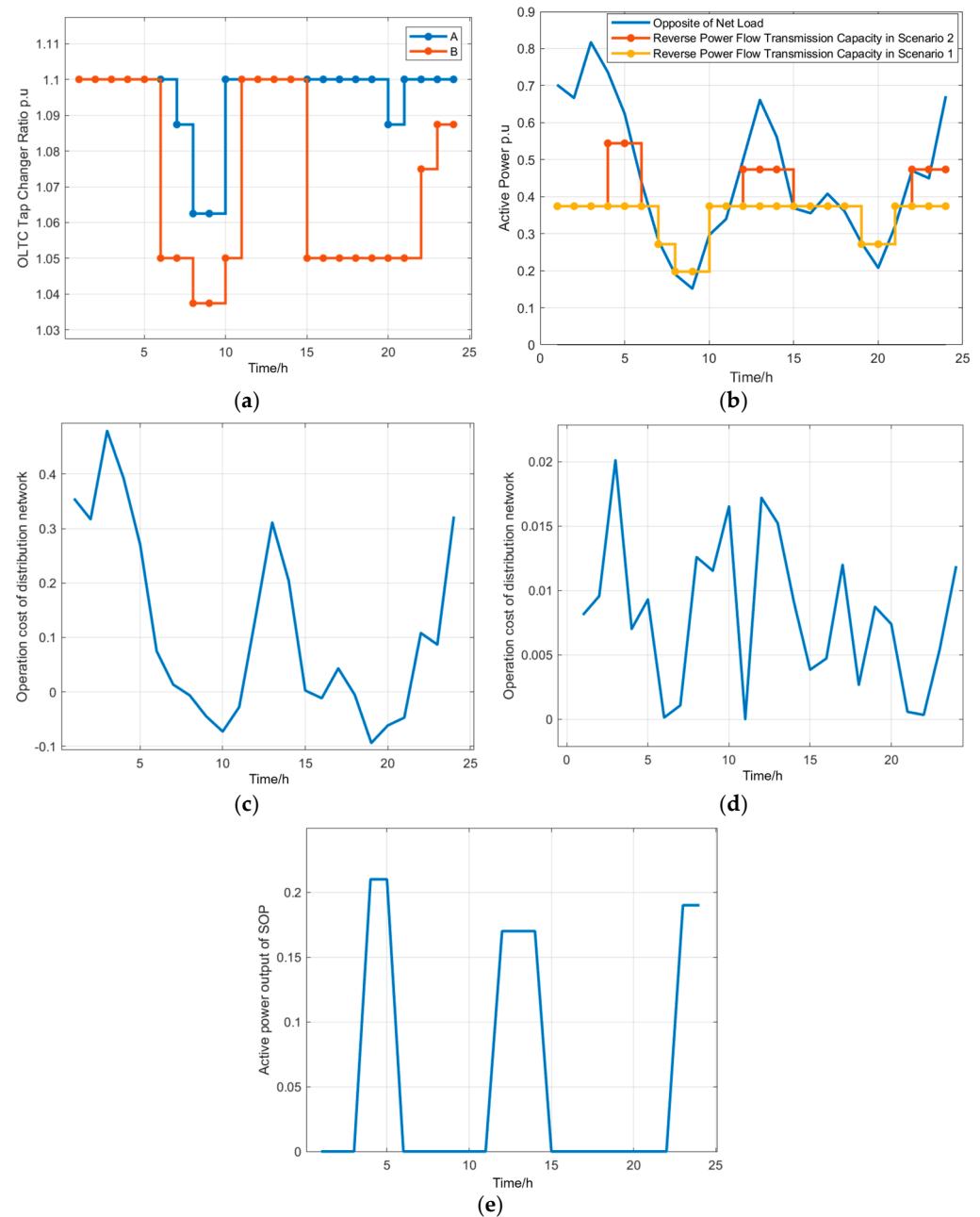


Figure 16. Joint operation of distribution networks A and B. (a) Depicts the OLTC tap change actions for distribution networks A and B. Compared to Scenario 1, the tap ratio for network A remains unchanged, whereas for network B, the ratio increases during periods 4–5, 12–14, and 22–24, leading to an increased reverse power flow capacity, which can be delivered through the SOP to network A. (b) Illustrates the curtailment of wind and solar power in network A during these periods, showing improvements during segments 4–5, 12–14, and 22–24. (c,d) Show the changes in operational costs

for networks A and B. The additional reverse power flow carried by network B results in increased network losses and higher operational costs; however, the improvement in curtailment for network A, when evaluated together with network B, leads to an overall reduction in total operational costs. (e) Output performance of the SOP.

5.2.3. Optimization Results of the Joint Operation of Distribution Networks A, B, and C

In Scenario 3, after the coordinated optimization of distribution networks A, B, and C through the SOP, the results improved significantly. As shown in Figure 17, the curtailment of wind and solar power in distribution network A improved.

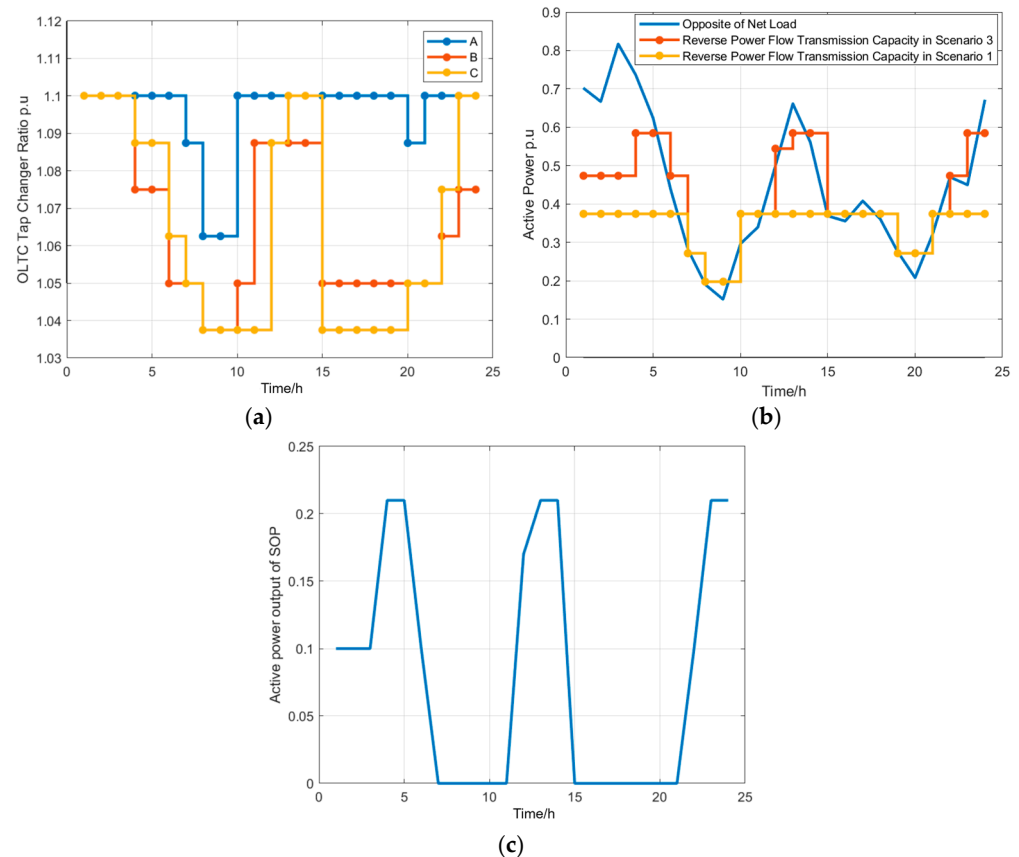


Figure 17. Joint operation of distribution networks A, B, and C. (a) The OLTC tap changes of networks A, B, and C show that taps of networks A and B remain unchanged, while network C increases its taps to transfer the reverse power flow from network A. This decision considers that network B still needs to handle its own reverse power flow, and network C's has the smallest, allowing more margin for transfer. (b) Depicts the curtailment of wind and solar in network A. After joint operation with networks B and C, there is a significant reduction in curtailment compared to Scenario 1 and 2. (c) Illustrates the SOP output at this time. Notably, during intervals like period 4, although OLTC taps in network B could be further increased to enhance reverse power flow capacity, the SOP has already reached its capacity limit. During intervals such as 1–3, while the SOP still has a margin, networks B and C also need to transfer reverse power flow to the main grid, leaving limited OLTC adjustment space, restricting the mitigation of reverse power flow for network A.

6. Conclusions

This paper investigates the challenges posed by the increasing penetration of distributed generation (DG) in distribution networks, particularly concerning the management of reverse power flow. In Section 2, an analysis is conducted on three factors affecting reverse power flow in distribution networks: the parameters of the on-load tap changer (OLTC), the reactive power compensation capacity installed on both sides of the OLTC, and the electrical distance from the DG to the OLTC. The following conclusions are drawn:

- (1) When transmitting reverse power flow, the OLTC, designed as a step-down transformer, is utilized as a step-up transformer. Consequently, even when the tap ratio of the OLTC reaches its limit, the ideal reverse power flow transmitted is only about 80% of its rated capacity. The specific impact of the OLTC tap ratio on the capacity for reverse power flow transmission is detailed in Equations (23) and (24) and Figure 6.
- (2) The reactive power compensation capacity on both sides of the OLTC also influences the transmission of reverse power flow. Without additional reactive power compensation, insufficient reactive power capacity on the low-voltage side of the OLTC restricts the transmission of reverse power flow to approximately 30% of the rated capacity. However, by adding extra reactive power compensation on the low-voltage side, the capacity for reverse power flow transmission can be increased to about 80% of the OLTC's rated capacity, albeit at the cost of additional installation, which raises the operational expenses of the distribution network.
- (3) The electrical distance from the DG to the OLTC further limits the capacity for reverse power flow transmission, which can drop to around 30% of the OLTC's capacity under the most adverse operating conditions.

Section 3 synthesizes the analysis of these three influencing factors and concludes that the impact of the OLTC tap ratio on reverse power flow transmission capacity is contradictory to the effect of the electrical distance from the DG to the OLTC. To address this, a parameter, b_t is introduced to describe the voltage on the low-voltage side of the OLTC. This, combined with the aforementioned factors, leads to a precise method for calculating the capacity for reverse power flow transmission in distribution networks, as outlined in Table 1.

Section 4 establishes a joint operational optimization model for multiple distribution networks that includes an SOP to enhance the reverse power flow transmission capability of individual distribution networks. In the case studies presented in Section 5, the operational results of the distribution networks are analyzed under independent operation, dual network operation, and triple network operation, leading to the following conclusions:

When the distribution network operates independently, the capacity for reverse power flow is relatively poor, leading to significant curtailment of wind and solar energy when distributed generation (DG) output is high. However, interconnecting multiple distribution networks through the SOP can facilitate the sharing of reverse power flow capabilities among different networks, thereby reducing the curtailment of wind and solar energy and enhancing overall operational efficiency. Nonetheless, this approach requires further refinement, as joint operation largely depends on the operational conditions of the various distribution networks. If multiple distribution networks simultaneously need to export reverse power flow, the effectiveness of this method will be constrained. Additionally, the high costs associated with the SOP significantly limit the output capacity for reverse power flow.

Future research should consider the integration of energy storage systems (ESS) to form an E-SOP, which would allow for the storage of electricity during peak DG output periods and enable export during times when the joint system has surplus capacity for reverse power flow. This will be a focal point for the next phase of research.

Author Contributions: Conceptualization, Z.M.; methodology, Z.M., F.C.; formal analysis, Z.M.; investigation, Z.M.; writing—original draft preparation, Z.M.; writing—review and editing, Z.M. and F.C.; funding acquisition, F.C. All authors have read and agreed to the published version of the manuscript.

Funding: This project is supported by the State Grid Science and Technology Project—Research And Application of Price Mechanism Theoretical Model for “Source-Network-Load-Storage Integration” at Park Level (1400-202457287A-1-1-ZN).

Institutional Review Board Statement: Not applicable.

Data Availability Statement: The original contributions presented in the study are included in the article; further inquiries can be directed to the corresponding authors.

Conflicts of Interest: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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